

Paper 8.25 - Toolkit for the Lattice Closure Template

CQE-paper-08.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-08.25.md

Paper 8.25 - Toolkit for the Lattice Closure Template

Purpose

Paper 8.25 describes the tools for reviewing the local lattice closure template. These tools expose the code/lattice checks and their audit boundaries; they do not close Leech or Gamma72 landing claims.

Review Objects

The toolkit works with:

```
chain rung          = 1, 3, 7, 8, 24
powered terminal    = 72 = 8 x 9
Fano/Hamming check  = dimension 7 incidence
extended Hamming check = dimension 8 E8 seed
Golay ingredient check = dimension 24 carrier
Gamma72 transport check = three-sheet round trip
audit boundary      = unproved landing or uniqueness claim
```

Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-08/verify_lattice_closure_template.py
production/formal-papers/CQE-paper-08/lattice_closure_template_receipt.json
```

The verifier imports:

```
lattice_forge.lattice_codes.verify_lattice_code_chain
lattice_forge.lattice_codes.verify_hamming_7_fano
lattice_forge.lattice_codes.verify_extended_hamming_8
lattice_forge.lattice_codes.verify_golay_24
lattice_forge.lattice_codes.verify_powered_chain
lattice_forge.nebe_gamma72.verify_nebe_gamma72_contract
```

Additional source and kernel files:

```
production/papers/CQE-paper-08/SOURCE.md
production/papers/CQE-paper-08/01-CQE-formal/FORMAL.md
production/papers/CQE-paper-08/02-CQE-tool/TOOL.md
production/papers/CQE-paper-08/03-CQE-workbook/WORKBOOK.md
production/paper-kernels/papers/CQE-paper-08/PAPER_KERNEL.md
```

The kernel notes currently mark `production/papers/CQE-paper-08/02-CQE-tool/run.py` as missing, so the promoted verifier is the replayable route for this paper.

Analog Toolkit

A physical reconstruction can be done as a rung table:

```
write the rungs 1, 3, 7, 8, 24, 72
record the certified fact for each rung
mark which facts are local and verified
mark which landings are open audit boundaries
draw 24 as three blocks of 8
draw 72 as eight sheets of nine or nine copies of eight
```

The purpose is not to reproduce all code theory by hand. The purpose is to make the proof boundary visible.

Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

```
What is the powered terminal of 8 x 9?  
Does Golay ingredient evidence prove the rootless Leech landing?  
Does Gamma72 round-trip transport prove the polarization action?  
What is the sheet K bound?
```

Expected answers:

```
72  
no  
no  
9
```

Boundary

Paper 8.25 is a toolkit supplement. Any stronger Leech, Gamma72, or uniqueness claim must pass Paper 8.50's claim contract before it can change the scientific paper.